Let N be the population size. Let τ be dimensionless time measured in units of the mean infectious period. Let $x(\tau)$ be the susceptible fraction, $y_1(\tau)$ the infected fraction of strain 1, and $y_2(\tau)$ the infected fraction of strain 2. Let ε be the demographic timescale parameter. Let $\mathcal{R}_{0,1}$ and $\mathcal{R}_{0,2}$ be the basic reproduction numbers of strain 1 and strain 2.

The deterministic two-strain system is

$$\frac{dx}{d\tau} = \varepsilon(1-x) - \mathcal{R}_{0,1}xy_1 - \mathcal{R}_{0,2}xy_2,$$

$$\frac{dy_1}{d\tau} = (\mathcal{R}_{0,1}x - 1)y_1,$$

$$\frac{dy_2}{d\tau} = (\mathcal{R}_{0,2}x - 1)y_2.$$

where

$$\mathcal{R}_{0,1} > 1, \qquad \mathcal{R}_{0,2} > 1.$$

The strain-specific boundary-layer thresholds are defined by

$$y_1^\star = \varepsilon\left(1 - \frac{1}{\mathcal{R}_{0,1}}\right),$$

$$y_2^\star = \varepsilon\left(1 - \frac{1}{\mathcal{R}_{0,2}}\right).$$

Assume strain 1 enters its boundary layer before strain 2. The opposite case is obtained by interchanging the labels 1 and 2.

Let τ_1 be the time at which strain 1 enters its boundary layer on the declining side of the epidemic wave. Define

$$x_a = x(\tau_1),$$

$$y_1(\tau_1) = y_1^\star,$$

$$y_{2,a} = y_2(\tau_1).$$

By assumption, strain 2 has not yet entered its own boundary layer at this time, so

$$y_{2,a} > y_2^\star.$$

The number of strain-1 infectives at this boundary-layer entrance is

$$m_1 = Ny_1^\star.$$

Define local time after strain 1 enters its boundary layer by

$$s = \tau - \tau_1.$$

Thus, at $s = 0$,

$$x(0) = x_a,$$

$$y_2(0) = y_{2,a}.$$

During the first stage, strain 1 is rare while strain 2 remains in the deterministic regime. We approximate strain 1 by a branching process and strain 2 by a deterministic trajectory. In this stage we ignore the feedback of strain 1 on the susceptible pool, and compared with the strain-2 transmission term, $\varepsilon(1-x)$ can be ignored during this phase. Thus the resident strain-2 trajectory is approximated by

$$\frac{dx}{ds} = -\mathcal{R}_{0,2} x y_2,$$
$$\frac{dy_2}{ds} = (\mathcal{R}_{0,2} x - 1) y_2.$$

By the chain rule,

$$\frac{dy_2}{dx} = \frac{dy_2/ds}{dx/ds}.$$

Substituting the two equations above gives

$$\frac{dy_2}{dx} = \frac{(\mathcal{R}_{0,2} x - 1) y_2}{-\mathcal{R}_{0,2} x y_2}.$$

Cancelling y_2,

$$\frac{dy_2}{dx} = -1 + \frac{1}{\mathcal{R}_{0,2} x}.$$

Integrating from $(x_a, y_{2,a})$ to $(x, y_2(x))$, using u as the integration variable,

$$y_2(x) - y_{2,a} = \int_{x_a}^{x} \left(-1 + \frac{1}{\mathcal{R}_{0,2} u} \right) du.$$

Therefore

$$y_2(x) = y_{2,a} + x_a - x + \frac{1}{\mathcal{R}_{0,2}} \ln \frac{x}{x_a}.$$

Let x_b be the susceptible fraction at which strain 2 enters its boundary layer. It is determined by

$$y_2(x_b) = y_2^\star.$$

Substituting the expression for $y_2(x)$,

$$y_{2,a} + x_a - x_b + \frac{1}{\mathcal{R}_{0,2}} \ln \frac{x_b}{x_a} = y_2^\star.$$

Define

$$C_2 = y_{2,a} + x_a - y_2^\star.$$

Then x_b satisfies

$$x_b - \frac{1}{\mathcal{R}_{0,2}} \ln \frac{x_b}{x_a} = C_2.$$

Equivalently, using the principal branch W_0 of the Lambert W function,

$$x_b = -\frac{1}{\mathcal{R}_{0,2}} W_0 \left(-\mathcal{R}_{0,2} x_a \exp(-\mathcal{R}_{0,2} C_2) \right).$$

At this point, strain 2 has

$$m_2 = N y_2^\star$$

infectives.

We now describe the distribution of strain 1 at the time strain 2 enters its boundary layer. Let $Z_1(s)$ be the number of strain-1 infectives during the first stage. Since strain 1 is rare, we approximate $Z_1(s)$ by a time-inhomogeneous linear birth-death process. At local time s, the per capita birth rate of strain 1 is

$$b_1(s) = \mathcal{R}_{0,1} x(s),$$

and the per capita death rate is

$$d_1(s) = 1.$$

For a process starting from one strain-1 infective, define the probability generating function

$$G_1(s; z) = \mathbb{E}[z^{Z_1(s)} \mid Z_1(0) = 1],$$

where z is the generating-function variable.

For a time-inhomogeneous linear birth-death process with per capita birth rate $b(s)$ and death rate $d(s)$, the probability generating function satisfies the backward equation

$$\frac{\partial G}{\partial s} = b(s)G^2 - [b(s) + d(s)]G + d(s),$$

with initial condition

$$G(0; z) = z.$$

In our case,

$$\frac{\partial G_1}{\partial s} = \mathcal{R}_{0,1}x(s)G_1^2 - [\mathcal{R}_{0,1}x(s) + 1]G_1 + 1,$$

with

$$G_1(0; z) = z.$$

This is the Riccati equation. Define the cumulative net growth

$$H_1(s) = \int_0^s [\mathcal{R}_{0,1}x(u) - 1]du,$$

where u is an integration variable. Define

$$A_1(s) = \int_0^s \mathcal{R}_{0,1}x(v)\exp(H_1(v))dv,$$

where v is an integration variable. Then the solution of the Riccati equation is

$$G_1(s; z) = 1 - \frac{\exp(H_1(s))(1 - z)}{1 + (1 - z)A_1(s)}.$$

We now rewrite these quantities in terms of x, avoiding the need to compute the time at which strain 2 enters its boundary layer. From

$$\frac{dx}{ds} = -\mathcal{R}_{0,2}xy_2(x),$$

we have

$$ds = -\frac{dx}{\mathcal{R}_{0,2}xy_2(x)}.$$

When the system moves from x_a to a smaller value u, define

$$\widehat{H}_1(u) = \int_u^{x_a} \frac{\mathcal{R}_{0,1}v - 1}{\mathcal{R}_{0,2}v\, y_2(v)}\, dv,$$

where v is an integration variable. In particular,

$$\widehat{H}_1(x_b) = \int_{x_b}^{x_a} \frac{\mathcal{R}_{0,1}v - 1}{\mathcal{R}_{0,2}v\, y_2(v)}\, dv.$$

And define

$$\widehat{A}_1 = \int_{x_b}^{x_a} \frac{\mathcal{R}_{0,1}}{\mathcal{R}_{0,2}y_2(u)} \exp(\widehat{H}_1(u))\, du,$$

where u is an integration variable.

Therefore, from one strain-1 infective at $x = x_a$, the probability generating function for the number of strain-1 infectives when strain 2 enters its boundary layer is

$$G_1^{1|2}(z) = 1 - \frac{\exp(\widehat{H}_1(x_b))(1-z)}{1 + (1-z)\widehat{A}_1}.$$

The superscript $1 \mid 2$ indicates that strain 1 is rare while strain 2 is resident.

If there are m_1 strain-1 infectives when strain 1 first enters its boundary layer, the branching property gives the total generating function

$$\left[G_1^{1|2}(z)\right]^{m_1}.$$

Now consider the second stage where both strains are in their boundary layers. Let r be local time starting from the moment strain 2 enters its boundary layer. At $r = 0$,

$$x(0) = x_b.$$

In this common boundary layer, both strains are rare. We ignore the feedback of infections on the susceptible pool, so susceptible recovery is dominated by demographic replenishment only:

$$\frac{dx}{dr} = \varepsilon(1-x).$$

Solving this equation gives

$$x(r) = 1 - (1-x_b)\exp(-\varepsilon r).$$

In the common boundary layer, the two strains are approximated as conditionally independent birth-death processes in the same externally determined susceptible environment $x(r)$. For strain i, where $i = 1, 2$, the per capita birth rate is

$$b_i^C(r) = \mathcal{R}_{0,i}x(r),$$

and the per capita death rate is

$$d_i^C(r) = 1.$$

The superscript C denotes the common boundary layer.

Define the cumulative net growth for strain i in the common boundary layer by

$$H_i^C(r) = \int_0^r [\mathcal{R}_{0,i}x(u) - 1]du.$$

Since

$$x(u) = 1 - (1-x_b)\exp(-\varepsilon u),$$

we obtain

$$H_i^C(r) = (\mathcal{R}_{0,i} - 1)r - \frac{\mathcal{R}_{0,i}(1-x_b)}{\varepsilon}(1 - \exp(-\varepsilon r)).$$

And define

$$A_i^C(r) = \int_0^r \mathcal{R}_{0,i}x(v)\exp(H_i^C(v))dv.$$

The probability generating function at time r is

$$G_i^C(r;z) = 1 - \frac{\exp(H_i^C(r))(1-z)}{1+(1-z)A_i^C(r)}.$$

Therefore the eventual extinction probability for one strain-i infective in the common boundary layer is

$$q_i^C = \lim_{r\to\infty} G_i^C(r;0).$$

Equivalently,

$$q_i^C = 1 - \lim_{r\to\infty} \frac{\exp(H_i^C(r))}{1+A_i^C(r)}.$$

Then we can get the burnout probability for strain 2. Strain 2 enters its boundary layer with

$$m_2 = Ny_2^\star$$

infectives. Each infective lineage has eventual extinction probability q_2^C. Hence

$$Q_2 = (q_2^C)^{m_2}.$$

For strain-1, if there are n strain-1 infectives at that moment, the probability that all of them eventually go extinct in the common boundary layer is

$$(q_1^C)^n.$$

Here n is the random variable $Z_1(x_b)$. We take the expectation

$$\mathbb{E}[(q_1^C)^{Z_1(x_b)}].$$

By the definition of a probability generating function, this is

$$G_1^{1|2}(q_1^C)$$

for a single initial strain-1 infective at $x = x_a$. Since strain 1 enters its boundary layer with m_1 infectives, the strain-1 burnout probability for the whole wave is

$$Q_1 = \left[G_1^{1|2}(q_1^C)\right]^{m_1}.$$

Thus the one-wave burnout probabilities under the two-boundary approximation are

$$Q_1 = \left[G_1^{1|2}(q_1^C)\right]^{m_1},$$

$$Q_2 = (q_2^C)^{m_2}.$$